Hyperthermia

TE-04 | Toxicology & Environmental

Quick flow

01 Remove From Heat [EMT / AEMT / Paramedic]

Move to cool area, remove tight/excess clothing, obtain glucose; temperature must not delay care.

02 Classify Severity [EMT / AEMT / Paramedic]

Cramps • exhaustion • suspected heat stroke (heat exposure plus altered mental status; a temperature below 104°F does not exclude it).

03 Heat Stroke? [EMT / AEMT / Paramedic]

Begin immediate active cooling concurrent with resuscitation. Ice-water immersion is preferred; use tepid-water immersion or tarp-assisted ice-water cooling if unavailable.

04 Cool to below 102°F [EMT / AEMT / Paramedic]

Use rectal core temperature when feasible, without interrupting care. Stop active cooling once core temperature is below 102°F (39°C).

05 Support Perfusion [AEMT / Paramedic]

For hypotension, dehydration, or poor perfusion, give isotonic crystalloid in reassessed aliquots: adult 500 mL; pediatric 20 mL/kg. Do not use a routine fixed maximum.

06 Reassess + Transport [EMT / AEMT / Paramedic]

Airway, ECG, neurologic status, BP, and core temperature; notify destination.

Overview

- Heat exposure with altered mental status is heat stroke until proven otherwise and demands immediate aggressive cooling; do not wait for a temperature or exclude heat stroke because the initial reading is below 104°F.
- Rapid cooling is initiated concurrently with standard resuscitation and takes precedence over transport in exertional heat stroke when effective immersion is available.

Provider-Level Actions

Protect rescuers, stop exposure when safe, support ABCs, and involve Poison Control or Medical Control early.

EMT

- Move to a cool environment, remove excess clothing, check glucose, begin active cooling, and give oral fluid only for mild illness with intact mentation/swallowing.

AEMT

- Perform all EMT care plus IV/IO isotonic crystalloid for hypotension, dehydration, or poor perfusion, with reassessment after each aliquot.

Paramedic

- Perform all prior care plus cardiac monitoring and protocol treatment of seizures, dysrhythmia, shock, or airway failure.

Indications

- Heat cramps, heat exhaustion, heat stroke, or other hyperthermic emergency.

Medications

Isotonic crystalloid

Adult 500 mL IV/IO aliquots; pediatric 20 mL/kg IV/IO aliquots; repeat only after reassessment for hypotension, dehydration, or poor perfusion.

Warnings / precautions

- Do not delay cooling to obtain a temperature; heat exposure with altered mental status is heat stroke until proven otherwise.
- Use rectal core temperature when feasible, but do not interrupt active cooling, airway management, or transport to obtain it.
- Do not overcool; stop active cooling once core temperature is below 102°F (39°C).
- Antipyretics and dantrolene are not treatments for environmental or exertional heat stroke; use active external cooling.

Source and references

Claiborne County EMS approved protocol manual. | Poison Help national line: 1-800-222-1222. | Current Tennessee EMS scope of practice and Claiborne County standing orders control for all provider permissions.

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